

Math Areas Checklist

*A simple guide to break down the major areas of math into readable sections, with examples, labels, definitions, and notes.
You can print this out and check things off as you learn each part.*

Arithmetic

☐ **Addition** — $3 + 4 = 7$
Combining two numbers to get a total.

☐ **Subtraction** — $10 - 6 = 4$
Finding the difference between numbers.

☐ **Multiplication** — $5 \times 3 = 15$
Repeated addition of the same number.

☐ **Division** — $20 \div 4 = 5$
Splitting a number into equal parts.

☐ **Fractions** — $1/2 + 1/2 = 1$
Parts of a whole.

☐ **Decimals** — $0.5 \times 10 = 5$
Fractions written using place value.

☐ **Percentages** — $20\% \times 50 = 10$
Parts out of 100.

Date completed: _____

Pre-algebra

☐ **Equation** — $x + 3 = 7$
A math statement with an equal sign showing two sides are the same.

☐ **Expression** — $2x - 5$
Uses numbers, variables, and operations but has no equal sign. If you add an equal sign, it becomes an equation.

☐ **Linear equation** — $4x = 20$
Variable is raised only to the first power; graphs as a straight line.

☐ **Inequality** — $x > 3$
Shows greater than, less than, or not equal to another value.

☐ **Order of operations** — $3 + 2 \times 4 = 11$

Rules for which operations to do first.

Date completed: _____

Algebra

☐ **Exponential equation** — $3^2 = 9$

Any equation involving exponents. Example: "three squared equals nine."

☐ **Square root** — $\sqrt{16} = 4$

A number that multiplies by itself to make the original value. ($4 \times 4 = 16$)

☐ **Linear equation** — $y = 2x + 1$

Creates a straight line when graphed. Non-linear equations create curves like parabolas or S-shapes.

☐ **Slope (rise over run)**

Measures steepness: change in vertical (rise) over change in horizontal (run). Example: A(2, 3), B(5, 11), Rise = $11 - 3 = 8$, Run = $5 - 2 = 3$, Slope = $8/3$

☐ **Linear function** — $f(x) = x + 5$

Straight line; slope 1, y-intercept 5.

☐ **Quadratic expression** — $x^2 + 3x + 2$

Has x^2 ; graphs as a parabola.

Date completed: _____

Geometry

☐ **Right angle** — 90°

A perfect corner.

☐ **Arithmetic equation** — $4 + 4 + 4 + 4 = 16$

Repeated addition.

☐ **Multiplication** — $3 \times 5 = 15$

Basic product.

☐ **Powers** — $2 \times 2 \times 2 = 8$

Repeated multiplication of the same number.

☐ **Number sequence** — 3, 4, 5

A list of numbers in a pattern.

Date completed: _____

Pre-calculus

☐ **Trigonometric equation** — $\sin(30^\circ) = 0.5$

Uses trig functions to relate angles and sides.

☐ **Quadratic function** — $y = x^2$

Creates a parabola when graphed.

☐ **Logarithmic equation** — $\log_{10}(100) = 2$

Uses logs to express exponents ($10^2 = 100$).

☐ **Arithmetic sequence** — 2, 4, 6, 8

Each term increases by a constant amount.

Date completed: _____

Calculus

☐ **Limit** — $\lim_{x \rightarrow 0} (x^2) = 0$

Describes what a function approaches as x gets close to a value.

☐ **Derivative** — $d/dx (x^2) = 2x$

Measures rate of change or slope at each point.

☐ **Integral** — $\int 2x \, dx = x^2$

Antiderivative; used to accumulate quantities like area.

☐ **Rate of change** — $\Delta \text{distance} / \Delta \text{time}$

Difference quotient; the idea that leads to derivatives.

Date completed: _____